(a) JSON Intermediate Representation

```
{  "parameters" : {
     "outer_diameter"  : {"value" : 20, "range": [ 10,40 ]},
     "inner_diameter"  : {"value" : 10, "range": [ 5,20 ]},
     "thickness" : {"value" : 2, "range": [ 1,4]} },
   "geometry" : {  "base_shapes" : [
     { "id" :"outer" ,  "type" :"cylinder" ,  "radius" :"outer_diameter/2"  },
     { "id" :"hole" ,   "type" :"cylinder" ,  "radius" :"inner_diameter/2"  }],
   "operations" : [{ "type" :"difference" ,  "targets" :[ "outer","hole"  ]}]} }
```

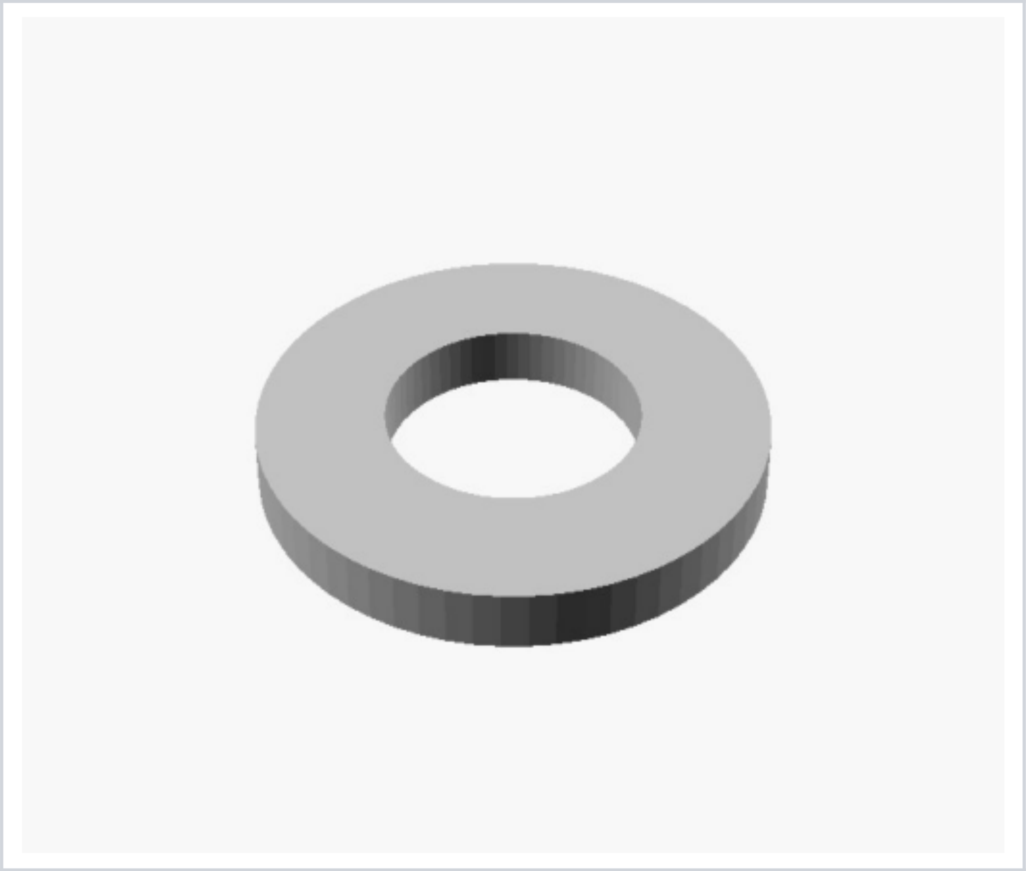
compile

(b) Generated OpenSCAD Code

```
outer_diameter =   20;  //[10:40:1]

inner_diameter =   10;  //[5:20:1]

thickness =  2;        //[1:4:0.5]

module washer() {
  difference() {
    cylinder(r= outer_diameter/2  , h=thickness);
    cylinder(r= inner_diameter/2  , h=thickness+2);
  }
}
```

OpenSCAD render

(c) Rendered Output



"outer_diameter/2" – preserved as expression, not evaluated to 10

"inner_diameter/2" – preserved as expression, not evaluated to 5